

Multi-Instrument Correlated Simulation Factor Model: Mathematical Foundations

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di.simbasket Module for KDB-X

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Abstract

This document presents the mathematical framework for generating correlated intraday price paths across a basket of instruments in the di.simbasket module. We adopt the Fama-French three-factor model to decompose each instrument's return into systematic factor exposures and an idiosyncratic residual. Factor paths are pre-generated as correlated Brownian motions and injected into the di.simtick price engine at each trade arrival, producing realistic intraday co-movement without sacrificing the microstructure realism of the underlying single-instrument simulator. Two approaches to volatility calibration are described: historical realised volatility and implied volatility extracted from options markets. Parameter estimation procedures are presented generically, applicable to any basket of equity instruments.

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1 Introduction

The `di.simtick` module generates realistic intraday tick data for a single instrument over a single trading day, combining a Hawkes self-exciting process for trade arrivals with Geometric Brownian Motion (GBM) or Merton jump-diffusion for price dynamics. `di.simcalendar` extends this to multiple trading days by chaining daily simulations and carrying forward closing prices.

`di.simbasket` introduces a further dimension: a portfolio of M instruments simulated simultaneously over N trading days. The key challenge is ensuring that price paths across instruments exhibit realistic co-movement. In real markets, correlated instruments react to the same macroeconomic shocks, sector news, and risk-off episodes. A naive approach — running `di.simtick` independently per instrument — produces uncorrelated paths that fail to capture this.

We address this through a factor model. Each instrument's return is decomposed into a small number of common factor exposures and an idiosyncratic residual. Factor paths are pre-generated as correlated Brownian motions and shared across all instruments. When `di.simtick` evaluates the price at each trade arrival, it samples from this shared factor path rather than drawing independent normal shocks — producing genuine intraday co-movement driven by common factors, while each instrument retains its own independent trade arrival process and idiosyncratic price noise.

We adopt the Fama-French three-factor model (Fama and French, 1993) as the factor structure. The three factors — market excess return, value (HML), and profitability (RMW) — are well-established, freely available from Kenneth French's data library, and provide sufficient explanatory power for a broad range of equity instruments.

2 Factor Model Framework

2.1 Model Specification

For instrument i at time t , the instantaneous return is decomposed as:

$$r_i(t) = \beta_{i,1} F_1(t) + \beta_{i,2} F_2(t) + \beta_{i,3} F_3(t) + \varepsilon_i(t)$$

where:

- $F_k(t)$ — return of common factor k at time t
- $\beta_{i,k}$ — sensitivity of instrument i to factor k (factor loading)
- $\varepsilon_i(t)$ — idiosyncratic residual, independent across instruments and factors

In continuous time, each factor $F_k(t)$ follows a Brownian motion with annualized volatility σ_k . Factors are correlated according to a $K \times K$ correlation matrix C . The idiosyncratic component $\varepsilon_i(t)$ follows an independent Brownian motion with volatility $\sigma_{\text{idio},i}$.

2.2 The Three Fama-French Factors

The Fama-French three-factor model decomposes equity returns into three systematic components, each capturing a distinct dimension of risk:

Factor	Symbol	Description	Economic interpretation
Market	Mkt-RF	Excess return of the market portfolio over the risk-free rate	Systematic market risk borne by all equities
Value	HML	Return of high book-to-market stocks minus low book-to-market stocks	Compensation for holding value vs. growth stocks
Profitability	RMW	Return of highly profitable firms minus weakly profitable firms	Compensation for holding firms with strong earnings

Table 1: Fama-French three factors used in `di.simbasket`.

2.3 Covariance Structure

The factor covariance matrix F is constructed from the factor volatility vector $\sigma = (\sigma_1, \dots, \sigma_K)$ and the factor correlation matrix C :

$$F = \text{diag}(\sigma) \cdot C \cdot \text{diag}(\sigma)$$

The total return covariance between instruments i and j is:

$$\text{Cov}(r_i, r_j) = B_i^T F B_j$$

where $B_i = (\beta_{i,1}, \dots, \beta_{i,K})^T$ is the vector of factor loadings for instrument i . The full $M \times M$ return covariance matrix for the basket is:

$$\Sigma = B F B^T + D$$

where B is the $M \times K$ matrix of factor loadings and $D = \text{diag}(\sigma_{\text{idio},1}^2, \dots, \sigma_{\text{idio},M}^2)$ is the diagonal matrix of idiosyncratic variances. The sum of two positive semi-definite matrices is always positive semi-definite, so Σ is guaranteed to be a valid covariance matrix by construction — a significant advantage over estimating a full $M \times M$ matrix directly, which requires $M(M-1)/2$ parameters and becomes unstable for large M .

2.4 Idiosyncratic Volatility

Given the total annualized volatility σ_i for instrument i and its factor loadings B_i , the idiosyncratic volatility is derived as:

$$\sigma_{\text{idio},i} = \sqrt{(\sigma_i^2 - B_i^T F B_i)}$$

This ensures that the total variance of each instrument exactly matches its calibrated volatility σ_i . A necessary condition is:

$$B_i^T F B_i < \sigma_i^2$$

If this condition is violated, the idiosyncratic variance is negative — mathematically inadmissible. The parameter estimation procedure includes a validation check, with beta shrinkage applied if necessary (see Section 6).

Note: The idiosyncratic volatility is never stored in basket.csv — it is computed at simulation load time from the stored total volatility and the factor structure.

3 Correlated Brownian Motion

3.1 Cholesky Decomposition

To generate correlated Brownian motions for the K factors, we apply the Cholesky decomposition of the factor correlation matrix C :

$$C = L L^T$$

where L is a lower triangular matrix with positive diagonal entries. Given K independent standard normal variates $Z = (Z_1, \dots, Z_K)^T$, the vector of correlated standard normals is:

$$W = L Z$$

The components of W satisfy $E[W_i W_j] = C_{ij}$, reproducing the desired correlation structure exactly.

3.2 Factor Path Pre-Generation

Factor paths are pre-generated on a fine time grid before any instrument simulation begins. Let T_{session} be the session duration in seconds and $\Delta t = 1$ second be the grid resolution, giving $n = T_{\text{session}} / \Delta t$ time steps.

For each time step $m = 1, \dots, n$:

- Draw K independent standard normals: $Z_m \sim N(0, I_K)$
- Compute correlated normals: $W_m = L Z_m$
- Scale to factor increments: $dF_{k,m} = \sigma_k \sqrt{(\Delta t / T_{\text{annual}})} W_{k,m}$
- Accumulate: $F_{k,m} = \sum_{j=1}^m dF_{k,j}$

where $T_{\text{annual}} = 252 \times T_{\text{session}}$ is the number of seconds in a trading year. The result is a $K \times n$ matrix of cumulative factor returns, shared across all instruments in the basket.

Note: Factor paths are regenerated independently for each trading day within the calendar loop — each day gets a fresh draw of correlated Brownian paths.

3.3 Factor Path Lookup at Trade Arrivals

When `di.simtick` generates a trade for instrument i at time t (seconds from session open), the cumulative factor contribution to the log-return since the previous trade at time t_{prev} is:

$$r_{\text{factor},i}(t) = \sum_{k=1}^K \beta_{i,k} [F_k(t) - F_k(t_{\text{prev}})]$$

The increment $F_k(t) - F_k(t_{\text{prev}})$ is obtained by linear interpolation between grid points. Since the grid resolution is 1 second and typical inter-trade times are also of order seconds, linear interpolation introduces negligible error.

The idiosyncratic contribution is drawn independently at each trade arrival:

$$r_{\text{idio},i}(t) = \sigma_{\text{idio},i} \sqrt{(\Delta t_i / T_{\text{annual}})} \varepsilon_i$$

where $\Delta t_i = t - t_{\text{prev}}$ and $\varepsilon_i \sim N(0,1)$. The total log-return driving the price update is:

$$r_i(t) = r_{\text{factor},i}(t) + r_{\text{idio},i}(t)$$

4 Price Path Decomposition

4.1 Modified GBM Under the Factor Model

In `di.simtick` without a factor model, the price at trade i is:

$$S_i = S_{i-1} \exp[(\mu - \sigma^2/2) \Delta t_i + \sigma \sqrt{\Delta t_i} \varepsilon_i]$$

Under the factor model, the normal shock $\sigma \sqrt{\Delta t_i} \varepsilon_i$ is replaced by the sum of factor and idiosyncratic components:

$$S_i = S_{i-1} \exp[(\mu - \sigma^2/2) \Delta t_i + r_{factor,i}(t_i) + r_{idio,i}(t_i)]$$

4.2 Variance Consistency

The total variance of the log-return over interval Δt is:

$$\begin{aligned} \text{Var}(r_i) &= \text{Var}(r_{factor,i}) + \text{Var}(r_{idio,i}) \\ &= (B_i^T F B_i + \sigma_{idio,i}^2) \Delta t / T_{annual} \\ &= \sigma_i^2 \Delta t / T_{annual} \end{aligned}$$

The last equality holds by construction of $\sigma_{idio,i}$. Each instrument's realised volatility matches its calibrated total volatility σ_i , regardless of the factor structure.

4.3 Backward Compatibility

The factor model is introduced via an optional factorpath key in the di.simtick configuration dictionary. When factorpath is null (ON in q), di.simtick uses its original independent normal draws. This preserves full backward compatibility: all existing simtick configurations, tests, and presets continue to work unchanged.

5 Parameter Estimation

5.1 Data Sources

Two freely available data sources are used:

- **Stock prices:** Daily adjusted closing prices, available from financial data providers (e.g. Yahoo Finance via the yfinance Python library). Adjustment for splits and dividends is applied automatically.
- **Factor returns:** Daily Fama-French five-factor returns from Kenneth French's data library (<https://mba.tuck.dartmouth.edu>). The five-factor daily file is used to access the RMW profitability factor, which is not present in the three-factor file.

The recommended estimation window is two years of daily data, providing approximately 500 observations — sufficient for stable OLS estimates while remaining recent enough to reflect current market conditions.

5.2 Factor Loadings: OLS Regression

For each instrument i , factor loadings are estimated by regressing daily excess returns on the three factor returns:

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_{i,1}(Mkt-RF)_t + \beta_{i,2} HML_t + \beta_{i,3} RMW_t + \varepsilon_{i,t}$$

where $r_{f,t}$ is the daily risk-free rate. The OLS estimator is:

$$B_i = (X^T X)^{-1} X^T y_i$$

where X is the $T \times 4$ matrix of factor returns with an intercept column and y_i is the $T \times 1$ vector of instrument excess returns. The intercept α_i (Jensen's alpha) is estimated but discarded — only the slope coefficients B_i are used.

The R^2 statistic measures the fraction of return variance explained by the three factors. Values below 0.3 may indicate an instrument with highly idiosyncratic behaviour, for which the factor model provides limited correlation structure.

5.3 Factor Volatility and Correlation

Factor annualized volatilities are estimated as:

$$\sigma_k = \text{std}(F_{k,1}, \dots, F_{k,T}) \times \sqrt{252}$$

The factor correlation matrix C is the sample Pearson correlation matrix of the K factor return series over the estimation window. Both σ_k and C are stored in the factor CSV files and remain fixed for all instruments in the basket.

5.4 Total Instrument Volatility: Two Approaches

The total annualized volatility σ_i can be calibrated from two sources, each with different forward-looking properties:

Approach	Source	Forward-looking	Typical use case
Historical realised vol	Daily closing prices (2-year window)	No — reflects the past	Backtesting, scenario generation
Implied volatility	ATM options prices (30-day horizon)	Yes — reflects market expectations	Stress testing, current-regime simulation

Table 2: Comparison of volatility calibration approaches.

5.5 Historical Realised Volatility

The historical annualized volatility for instrument i is:

$$\sigma_{hist,i} = std(r_{i,1}, \dots, r_{i,T}) \times \sqrt{252}$$

where $r_{i,t} = \ln(S_{i,t} / S_{i,t-1})$ are daily log-returns over the estimation window of T trading days. This is the standard close-to-close estimator.

Historical vol has the advantage of being easy to compute and consistent with the factor beta estimation window. Its main limitation is that it reflects the past regime and may be slow to react to changes in market conditions — for example, after a macro shock or earnings surprise.

5.6 Implied Volatility

Implied volatility (IV) is extracted from the prices of exchange-traded options via the Black-Scholes inversion. For each instrument i , we use the at-the-money (ATM) implied volatility from the nearest monthly expiry with approximately 30 days to expiration:

$$C_{BS}(S, K, r, \sigma_{IV}, T) = C_{market}$$

where C_{BS} is the Black-Scholes call price, $K \approx S$ is the ATM strike, and C_{market} is the observed option mid-price. The implied volatility $\sigma_{IV,i}$ is solved numerically (e.g. via Brent's method).

IV is forward-looking: it reflects the market's expectation of future volatility over the option's remaining life. It typically exceeds realised volatility by a few percent, a phenomenon known as the variance risk premium:

$$E[\sigma_{IV}] \approx \sigma_{realised} + VRP$$

where VRP is typically 2–5% for large-cap equities. For stress testing and forward-looking scenario generation, using IV directly is the natural choice — the premium is a feature, not a bug, as it produces conservatively volatile simulations.

If the goal is to match historical return distributions precisely, a scaling adjustment can be applied:

$$\sigma_i = \sigma_{IV,i} \times (\sigma_{hist,i} / \sigma_{IV,i})$$

though this reduces IV to the historical estimate and loses the forward-looking benefit.

Note: IV is not available for all instruments. Thinly traded options may have wide bid-ask spreads that distort the implied vol estimate. In such cases, historical vol is the safer choice.

5.7 Choosing Between the Two

The choice depends on the simulation objective:

- **Use historical vol** when the goal is to reproduce the statistical properties of a past period — e.g. backtesting a strategy against 2023 market conditions.
- **Use implied vol** when the goal is to simulate plausible future scenarios under current market expectations — e.g. stress testing a portfolio before an earnings season.
- **Mix the two** for specific instruments: use IV for liquid large-caps with active options markets, historical vol for less liquid names.

Regardless of which approach is used, the betas and factor structure are always estimated from historical data — the options market does not provide information about factor sensitivities.

5.8 Idiosyncratic Volatility

Once σ_i is chosen (historical or implied), the idiosyncratic volatility is derived as described in Section 2.4:

$$\sigma_{idio,i} = \sqrt{(\sigma_i^2 - B_i^T F B_i)}$$

This derivation is performed at simulation load time and is not stored. The value of σ_i stored in basket.csv is the only volatility parameter the user needs to specify.

Note: When using implied vol, verify that the idiosyncratic vol check passes (Section 6.1). High IV can sometimes cause $B_i^T F B_i$ to exceed σ_i^2 if betas were estimated in a low-vol historical period.

6 Validation

6.1 Idiosyncratic Volatility Check

Before any simulation, `di.simbasket` validates that the idiosyncratic variance is positive for each instrument:

$$\sigma_{idio,i}^2 = \sigma_i^2 - B_i^T F B_i > 0 \forall i$$

If this condition fails for instrument i , the factor loadings explain more variance than the instrument exhibits — an inadmissible configuration. Beta shrinkage is applied:

$$B_i \leftarrow s \cdot B_i \quad s = \sqrt{(0.9 \sigma_i^2 / B_i^T F B_i)}$$

This scales the loadings so that factor-explained variance equals 90% of total variance, leaving 10% idiosyncratic. The shrinkage factor s and the affected instruments are reported to the user.

6.2 Correlation Recovery Check

After simulation, the realised pairwise correlation between instruments i and j (computed from daily closing returns) should approximately match the theoretical correlation implied by the factor model:

$$\rho_{ij}^{theory} = (B_i^T F B_j) / (\sigma_i \sigma_j)$$

Agreement within sampling noise (approximately ± 0.05 for 252-day simulations) confirms the factor injection is working correctly. Systematic deviations indicate a bug in the path lookup or idiosyncratic scaling.

6.3 Volatility Recovery Check

The annualized realised volatility of each instrument should match its calibrated σ_i :

$$\sigma_{realised,i} = std(r_{i,1}, \dots, r_{i,N}) \times \sqrt{252} \approx \sigma_i$$

where $r_{i,d}$ is the daily log-return on day d . Significant deviations indicate an error in the idiosyncratic volatility scaling.

6.4 Factor Path Correlation Check

The pre-generated factor paths should exhibit the target correlation structure. For a simulation of sufficient length, the sample correlation between factor paths k and l should satisfy:

$$corr(F_k, F_l) \approx C_{kl}$$

This check can be performed on the pre-generated paths before any instrument simulation, providing an early warning of errors in the Cholesky decomposition step.

7 Integration with `di.simtick`

7.1 Configuration Extension

The factor model is introduced via a single optional key in the `di.simtick` configuration dictionary:

```
cfg[`factorpath`]: (times; factor_matrix; betas; idiovol)
```

where:

- **times** — n -element vector of timestamps (seconds from session open)
- **factor_matrix** — $K \times n$ matrix of cumulative factor returns
- **betas** — K -element vector of factor loadings for this instrument

– **idiovol** — scalar idiosyncratic annualized volatility

When `factorpath` is null (0N), `di.simtick` behaves exactly as in the standalone configuration — drawing independent normal shocks. This preserves full backward compatibility.

7.2 Modified Price Generation

```
/ Standalone simtick: independent normal draw shocks: vol * sqrt[dtyears] *  
normal[n; cfg] / With factor model: decompose into factor + idiosyncratic  
factor_return: interpolate factorpath at each trade time idio_shock: idiovol  
* sqrt[dtyears] * normal[n; cfg] shocks: factor_return + idio_shock
```

7.3 simbasket Orchestration

```
1. Load basket.csv -> per-instrument cfg dicts 2. Load factors.csv -> factor  
correlation matrix C 3. Load factorvols.csv -> factor vols sigma 4. Compute  $F$   
 $= \text{diag}(\text{sigma}) * C * \text{diag}(\text{sigma})$  5. Compute idiovol(i) for each instrument  
[validate > 0] 6. For each trading day d: a. Pre-generate factor paths  $F_k(t)$   
via Cholesky b. For each instrument i: - Inject factorpath into cfg - Call  
simtick.run[cfg] [simcalendar handles day loop] 7. Collect and return all  
trade/quote tables
```

Steps 1-5 are performed once at startup. The factor path pre-generation (step 6a) runs once per trading day. The instrument simulations (step 6b) are independent and can be parallelised across instruments using `peach` on multi-core KDB-X licences.

8 References

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